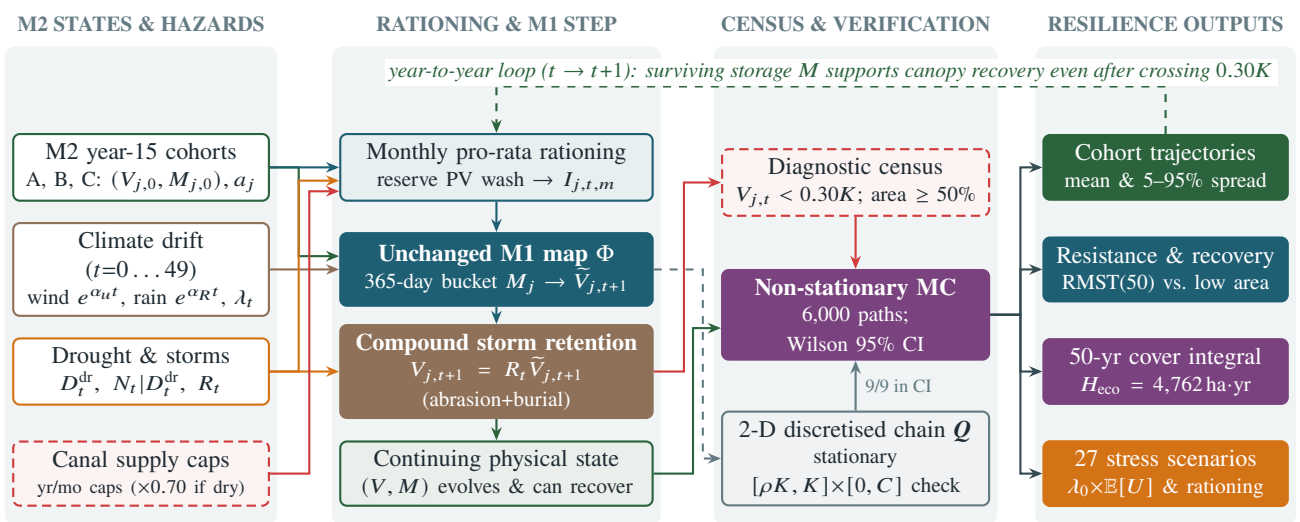

M2 STATES & HAZARDS
RATIONING & M1 STEP
CENSUS & VERIFICATION
RESILIENCE OUTPUTS
year-to-year loop ($t \to t+1$): surviving storage M supports canopy recovery even after crossing $0.30K$
M2 year-15 cohorts
A, B, C: $(V_{j,0}, M_{j,0}), a_j$
Climate drift
(t=0 . . . 49)
wind $e^{\alpha_u t}$, rain $e^{\alpha_R t}$, λ_t
Drought & storms
D_t^{dr}, $N_t | D_t^{\mathrm{dr}}$, R_t
Canal supply caps
yr/mo caps (×0.70 if dry)
Monthly pro-rata rationing
reserve PV wash → $I_{j,t,m}$
Unchanged M1 map Φ
365-day bucket $M_j \to \widetilde{V}_{j,t+1}$
Compound storm retention
$V_{j,t+1} = R_t \widetilde{V}_{j,t+1}$
(abrasion+burial)
Continuing physical state
(V, M) evolves & can recover
Diagnostic census
$V_{j,t} < 0.30K$; area ≥ 50%
Non-stationary MC
6,000 paths;
Wilson 95% CI
9/9 in CI
2-D discretised chain Q
stationary
$[\rho K, K] \times [0, C]$ check
Cohort trajectories
mean & 5–95% spread
Resistance & recovery
RMST(50) vs. low area
50-yr cover integral
$H_{\mathrm{eco}} = 4{,}762$ ha·yr
27 stress scenarios
$\lambda_0 \times \mathbb{E}[U]$ & rationing